GreenRoute: An AI Advisor for Lower-Emission Freight Routes

ELEC5623 Group Business Proposal · Track A — AI Agent Product

Group 25

Cover information	
Track	A — AI Agent Product
Group	25
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Product vision	Help freight companies cut emissions with AI.

1. Executive Summary

Transport is a major source of greenhouse gas emissions. GreenRoute is an AI agent that helps freight companies choose routes with lower emissions. The user enters a delivery job and GreenRoute suggests a greener route and explains why. This helps companies meet sustainability goals and save fuel.

2. Background

Freight companies plan many deliveries every day. Climate change is an important issue and companies are under pressure to reduce emissions. AI can help with many problems including logistics.

3. Problem Statement

Freight companies do not know which routes have the lowest emissions. GenAI can explain route choices in natural language so that dispatchers understand them.

4. Requirements

Users are dispatchers at freight companies.

ID	**Requirement**
FR1	User can enter a delivery job
FR2	System suggests routes
FR3	System estimates emissions for each route
FR4	System explains the suggestion
FR5	User can compare routes
NFR1	System should be fast
NFR2	System should be accurate
NFR3	System should be user-friendly

5. Proposed Product

GreenRoute is a web app. The dispatcher enters the origin, destination and vehicle type. The system gets candidate routes from a routing API, estimates the emissions of each route from distance, vehicle type and expected traffic, and then an LLM agent compares the routes and writes a recommendation with an explanation. The agent can also answer follow-up questions such as "what if we leave two hours later?" by calling the routing tool again.

6. Methodology

We will use Python, a routing API such as OpenRouteService, and an LLM API. Emissions will be estimated with a simple formula based on fuel consumption per kilometre for each vehicle type. We will use an agile approach.

7. Business Analysis

GreenRoute saves fuel and reduces emissions, which is good for companies and the environment. Companies could pay a monthly subscription.

8. Evaluation

We will test GreenRoute on some example delivery jobs and check that the recommendations make sense. We will also ask some people to try it.

9. Risks

The emissions estimates may not be accurate. The routing API may have limits.

10. Plan

Weeks	Work
5–8	Build the system
9–11	Test and improve
12–13	Final report and presentation

11. References

International Energy Agency. (2023). *Tracking clean energy progress 2023*. IEA.

OpenRouteService. https://openrouteservice.org